

Cauchy-characteristic matching for the Z4c evolution system: method

z4c-CMM verified research loop¹

¹generated from the mission-2 knowledge ledger (*knowledge-database/paper_z4c-CCM/*);
every equation and number below is traceable to a verifier output under *results/numerical/*
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I. INTRODUCTION

Cauchy-characteristic matching (CCM) couples a strongly hyperbolic interior (Cauchy) evolution to a characteristic evolution reaching future null infinity, so that the worldtube supplying characteristic boundary data simultaneously receives the incoming radiative content computed on the characteristic grid; the interface becomes transparent rather than approximately absorbing [? ?]. The present paper formulates and verifies CCM for the Z4c formulation of the Einstein equations [? ? ?]: the constraint-damped conformal decomposition in production use across finite-difference numerical-relativity codes, here the AthenaK implementation [?]. The presentation follows the organization of Ref. [?]: the Cauchy system and its boundary sector (Sec. II), the characteristic system (Sec. III), the matching algorithm (Sec. IV), and numerical verification (Sec. V). Every equation is one of: (a) a machine-verified result of this program (dark blue; verification labels [Vn] and the ledger nodes N1–N16); (b) standard literature content, cited at the equation; (c) an explicitly open obligation (labels O-Nx-y). The derivation chain is summarized as a directed acyclic graph in Fig. ??.

II. THE Z4C CAUCHY SYSTEM AND ITS BOUNDARY SECTOR

Black equations are standard literature content, cited at the equation; **dark blue** equations are the machine-verified new Z4c-CCM content (verification labels [Vn], Table II; ledger nodes in section titles).

A. Boundary frame and Z4c variables

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad n^\mu = \alpha^{-1} (1, -\beta^i), \quad (1)$$

$$s_i = \frac{\partial_i r}{\sqrt{\gamma^{rr}}}, \quad s^i = \frac{\gamma^{ir}}{\sqrt{\gamma^{rr}}}, \quad m^A = \frac{1}{\sqrt{2}} (e_2 + i e_3)^A, \quad (2)$$

$$\mathring{l}^\mu = \frac{1}{\sqrt{2}} (n^\mu + s^\mu), \quad \mathring{k}^\mu = \frac{1}{\sqrt{2}} (n^\mu - s^\mu) \quad [?]. \quad (3)$$

Z4c variables [?]:

$$\chi = (\det \gamma)^{-1/3}, \quad \tilde{\gamma}_{ij} = \chi \gamma_{ij}, \quad K = \hat{K} + 2\Theta, \quad \tilde{A}_{ij} = \chi \left(K_{ij} - \frac{1}{3} \gamma_{ij} K \right), \quad (4)$$

$$\partial_t \chi = \frac{2}{3} \chi [\alpha (\hat{K} + 2\Theta) - D_i \beta^i], \quad (5)$$

$$\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + \beta^k \tilde{\gamma}_{ij,k} + 2\tilde{\gamma}_{k(i} \beta_{j)}^k - \frac{2}{3} \tilde{\gamma}_{ij} \beta^k_{,k} \quad [?]. \quad (6)$$

B. The Z4c evolution system

The Z4c formulation [? ?] evolves the conformal variables (4) together with $\hat{K} = K - 2\Theta$, the conformal trace-free curvature $\tilde{A}_{ij}$, the constraint pair (Θ, Z_i) — with Z_i carried by $\tilde{\Gamma}^i = \tilde{\gamma}^{jk} \tilde{\Gamma}_{jk}^i + 2\tilde{\gamma}^{ij} Z_j$ — and the gauge (α, β^i) . In vacuum, with D_i the γ -covariant derivative and $[\cdot]^\text{TF}$ the γ -trace-free part, the evolution equations are [?]

$$\partial_t \hat{K} = \beta^i \partial_i \hat{K} - D^i D_i \alpha + \alpha \left[\tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3} (\hat{K} + 2\Theta)^2 \right] + \alpha \kappa_1 (1 - \kappa_2) \Theta, \quad (7)$$

$$\partial_t \tilde{A}_{ij} = \beta^k \partial_k \tilde{A}_{ij} + 2 \tilde{A}_{k(i} \partial_{j)} \beta^k - \frac{2}{3} \tilde{A}_{ij} \partial_k \beta^k + \chi [\alpha R_{ij} - D_i D_j \alpha]^{\text{TF}} + \alpha \left[(\hat{K} + 2\Theta) \tilde{A}_{ij} - 2 \tilde{A}_{ik} \tilde{A}^k_{\ j} \right], \quad (8)$$

$$\partial_t \Theta = \beta^i \partial_i \Theta + \frac{\alpha}{2} \left[\mathcal{R} + \frac{2}{3} (\hat{K} + 2\Theta)^2 - \tilde{A}_{ij} \tilde{A}^{ij} \right] - \alpha \kappa_1 (2 + \kappa_2) \Theta, \quad (9)$$

$$\begin{aligned} \partial_t \tilde{\Gamma}^i &= \beta^k \partial_k \tilde{\Gamma}^i - \tilde{\Gamma}^k \partial_k \beta^i + \frac{2}{3} \tilde{\Gamma}^i \partial_k \beta^k + \tilde{\gamma}^{jk} \partial_j \partial_k \beta^i + \frac{1}{3} \tilde{\gamma}^{ij} \partial_j \partial_k \beta^k \\ &\quad - 2 \tilde{A}^{ij} \partial_j \alpha + 2\alpha \left[\tilde{\Gamma}^i_{\ jk} \tilde{A}^{jk} - \frac{3}{2} \tilde{A}^{ij} \partial_j \ln \chi - \frac{1}{3} \tilde{\gamma}^{ij} \partial_j (2\hat{K} + \Theta) - \kappa_1 \tilde{\gamma}^{ij} Z_j \right], \end{aligned} \quad (10)$$

with $\mathcal{R}$ the Ricci scalar of γ_{ij} and (κ_1, κ_2) the constraint-damping parameters (production: $\kappa_1 = 0.02$, $\kappa_2 = 0$). The gauge is 1+log slicing and the gamma-driver shift [? ?],

$$\partial_t \alpha = \beta^i \partial_i \alpha - \mu_L \alpha^2 \hat{K}, \quad \partial_t \beta^i = \mu_S \tilde{\Gamma}^i - \eta \beta^i + \beta^j \partial_j \beta^i, \quad (11)$$

(AthenaK realization: $\mu_L = 2/\alpha$, $\mu_S = 1$, $\eta = 2$ [?]). The constraints are

$$\mathcal{H} = \mathcal{R} + \frac{2}{3} K^2 - \tilde{A}_{ij} \tilde{A}^{ij}, \quad \mathcal{M}_i = D^j (K_{ij} - \gamma_{ij} K), \quad \Theta = 0, \quad Z_i = 0, \quad (12)$$

and at the principal level the constraint subsystem propagates as waves, $\square \Theta \simeq 0 \simeq \square Z_i$ [?], the fact underlying the boundary rows of Sec. IIF and the stability analysis of node N13.

C. The ten incoming boundary modes

At a timelike outer boundary with frame (1)–(3), the linearized Z4c system carries ten incoming characteristic fields (taxonomy verified at nodes N4/N5: transverse-traceless = physical; h_{ss} = gauge; h_{sA} = constraint-vector; tangential trace = constraint-scalar): The constraint and gauge rows are those of Ref. [?] adapted to the Z4c variable set;

Table I. Incoming boundary modes of Z4c-CCM and their data sources (ledger node N6). Per-speed outgoing characteristic vectors $\hat{X}^a = (\hat{n}^a + \sqrt{v} \hat{s}^a)/\sqrt{2}$ at sector speed v [?]; production order $L \geq 1$.

#	sector	incoming field	speed	boundary condition (datum)
1–2	physical	$U_{AB}^- = \text{TT}_2[E + \epsilon(s)B]_{AB}$	1	$U_{AB}^- \hat{=} -(\psi'_0 \tilde{m}_A \tilde{m}_B + \text{c.c.})$ [CCM: $\psi'_0 = A^2 \psi_0^{\text{CCE}}$]
3	constraint	Θ	1	$(r^2 \hat{l} \cdot \partial)^{L+1} \Theta \hat{=} 0$
4	constraint	Z_s	1	$(r^2 \hat{l} \cdot \partial)^{L+1} Z_s \hat{=} 0$
5–6	constraint	Z_A	1	$(r^2 \hat{l} \cdot \partial)^{L+1} Z_A \hat{=} 0$
7	gauge	α	$\sqrt{\mu_L}$	$(r^2 \hat{m} \cdot \partial)^{L+1} \alpha \hat{=} h_\alpha$
8	gauge	β_s	$\sqrt{\nu_s}$	$(r^2 \hat{k}_{\nu_s} \cdot \partial)^{L+1} \beta_s \hat{=} h_s$
9–10	gauge	β_A	$\sqrt{\mu_S}$	$(r^2 \hat{j} \cdot \partial)^{L+1} \beta_A \hat{=} h_A$

the physical rows 1–2 carry the CCM datum and constitute the new content (Secs. IID–IIE, IIF). With $\psi'_0 \rightarrow 0$ the set reduces exactly to the absorbing constraint-preserving scheme of Ref. [?] [V6]. The frozen-coefficient LF/GKS analysis of the constraint rows (node N13) shows that every matched-falloff advection realization of rows 3–6 is stable and absorbs the (Θ, Z_s) pair at $1 - O((\sigma/2\omega r)^2)$, that falloff-mismatched sink variants self-generate $O(10\%)$ pair reflection, and that the production default (the stock matched realization) is the admissible optimum.

D. Tetrad correspondence and Type-II boost (node N2)

Outgoing null label u with $u|_{\partial\Omega} = t$: $\partial_\mu u = (1, u_r, 0, 0)$,

$$g^{tt} + 2g^{tr} u_r + g^{rr} u_r^2 = 0 \quad \implies \quad u_r = \frac{-g^{tr} - \sqrt{(g^{tr})^2 - g^{tt} g^{rr}}}{g^{rr}}, \quad u_r|_{\text{flat}} = -1. \quad (13)$$

Characteristic outgoing null vector [? ?]:

$$\hat{l}^\mu = -e^{2\hat{\beta}} g^{\mu\nu} \partial_\nu u. \quad (14)$$

For arbitrary $(\alpha, \beta^i, \gamma_{ij})$,

$$\hat{l}^\mu = \frac{\sqrt{2}}{2} (\alpha - \gamma_{ij} \beta^i \beta^j) e^{-2\hat{\beta}} \hat{l}^\mu \quad [\text{V1: exact at 12 rational points; GPU residual } 9.3 \times 10^{-16}], \quad (15)$$

$$A = (\alpha - \gamma_{ij}\beta^i s^j) e^{-2\hat{\beta}}, \quad \psi'_0 = A^2 \psi_0, \quad (16)$$

i.e. the Choice-2 boost of Ref. [?] holds verbatim on the Z4c frame (3): no evolution-system variable enters (15). ψ_0 on the characteristic grid [? ?]:

$$\psi_0 = \left(\frac{r\partial_r\beta - 1}{4Kr} \right) \left[(1+K)\partial_r J - \frac{J^2\partial_r\bar{J}}{1+K} \right] + \frac{J(1+K^2)\partial_r J\partial_r\bar{J}}{8K^3} + \frac{1}{8K} \left[\frac{J^2\partial_r^2\bar{J}}{1+K} - (1+K)\partial_r^2 J \right] - \frac{J\bar{J}^2(\partial_r J)^2 + J^3(\partial_r\bar{J})^2}{16K^3}. \quad (17)$$

E. Physical-mode dictionary (node N3, exact)

Electric/magnetic Weyl decomposition w.r.t. n^μ (vacuum):

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta} n^\alpha n^\beta, \quad B_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\alpha}{}^{\gamma\delta} C_{\gamma\delta\nu\beta} n^\alpha n^\beta, \quad \epsilon_{\mu\nu\rho\sigma} = \sqrt{-g} [\mu\nu\rho\sigma]. \quad (18)$$

EXACT identity (pure Weyl algebra, generic 10-component tensor; no perturbation, no field equations) [V2a]:

$$(E_{\mu\nu} + \epsilon_\mu{}^{\kappa\lambda} s_\kappa B_{\lambda\nu}) m^\mu m^\nu = -\psi_0, \quad (E_{\mu\nu} - \epsilon_\mu{}^{\kappa\lambda} s_\kappa B_{\lambda\nu}) m^\mu m^\nu = -\bar{\psi}_4, \quad (19)$$

with $\psi_0 = -C_{abcd} l^a m^b l^c m^d$ on the frame (3); dyad completeness ($U_{AB} = U(m, m) \bar{m}_A \bar{m}_B + \text{c.c.}$ for tangential trace-free U) gives the tensor form of the incoming Weyl object

$$U_{AB}^- \equiv \text{TT}_2[E + \epsilon(s)B]_{AB} = -(\psi_0 \bar{m}_A \bar{m}_B + \bar{\psi}_0 m_A m_B). \quad (20)$$

Realization from the Z4c state (Gauss–Codazzi, on-shell vacuum; γ_{ij}, K_{ij} from (4)) [V2c]:

$$E_{ij} = R_{ij}[\gamma] + K K_{ij} - K_{ik} K^k{}_j, \quad B_{ij} = \epsilon_i{}^{kl} D_k K_{lj}|_{\text{sym}}, \quad \epsilon_{ikl} = \sqrt{\det \gamma} [ikl]. \quad (21)$$

Nonperturbative verification (Kerr–Schild Schwarzschild, autodiff, both routes; Table II V2c): $|E^{3+1} - E^{4D}|$ rel. 7.7×10^{-17} , $|B^{3+1} - B^{4D}|$ rel. 3.9×10^{-17} , identity residual 8.5×10^{-16} . Linearized on-shell flat limit (superseded defining form, now a corollary [V2b]): for TT $h_{AB}(t, x_s)$, $\psi_0 = \frac{1}{4}(\partial_t + \partial_s)^2 h_{mm}$ and $U_{AB}^- \rightarrow -\frac{1}{4}(\partial_t + \partial_s)^2 \gamma_{AB}^{\text{TF}}$, i.e. $(\partial_t + \partial_s)^2 \gamma_{AB}^{\text{TF}} = 4(\psi_0 \bar{m}_A \bar{m}_B + \text{c.c.})$.

F. The Z4c-CCM boundary conditions (node N6)

Ten incoming modes at the worldtube; $X_v^\mu = (n^\mu + \sqrt{v} s^\mu)/\sqrt{2}$ at sector speed v ; $L \geq 1$:

$$\text{physical (2):} \quad U_{AB}^- \hat{=} -(\psi'_0 \bar{m}_A \bar{m}_B + \bar{\psi}'_0 m_A m_B), \quad \psi'_0 = A^2 \psi_0^{\text{CCE}} \quad (\text{exact, (20)–(21)}), \quad (22)$$

$$\text{constraint (4):} \quad (r^2 \mathring{l} \cdot \partial)^{L+1} \Theta \hat{=} 0, \quad (r^2 \mathring{l} \cdot \partial)^{L+1} Z_s \hat{=} 0, \quad (r^2 \mathring{l} \cdot \partial)^{L+1} Z_A \hat{=} 0, \quad (23)$$

$$\text{gauge (4):} \quad (r^2 \mathring{m} \cdot \partial)^{L+1} \alpha \hat{=} 0, \quad (r^2 \mathring{k}_{\nu_s} \cdot \partial)^{L+1} \beta_s \hat{=} 0, \quad (r^2 \mathring{j} \cdot \partial)^{L+1} \beta_A \hat{=} 0. \quad (24)$$

Björhus form of (22) (the correction drives the incoming Weyl object to its CCM value, the Z4c analog of Ref. [?]):

$$U_{AB}^- - U_{AB}^-|_{\text{BC}} \rightarrow 0, \quad U_{AB}^-|_{\text{BC}} = -(\psi'_0 \bar{m}_A \bar{m}_B + \bar{\psi}'_0 m_A m_B); \quad \psi_0^{\text{CCE}} \rightarrow 0 \implies \text{CPBC scheme of Ref. [?]} \quad [\text{V6}]. \quad (25)$$

Equations (23)–(24) are the order- L conditions of Ref. [?]; (24) replaces the Sommerfeld gauge placeholder of Ref. [?].

G. Sector orthogonality (nodes N4, N5)

For arbitrary TT data $h_{AB}(t, x_s)$ [V4]:

$$\mathcal{H} = \partial_i \partial_j h_{ij} - \partial^2 h^k{}_k = 0, \quad \mathcal{M}_i = \partial_j \dot{h}_{ij} - \partial_i \dot{h}^j{}_j = 0, \quad \gamma^{ij} \partial_t h_{ij} = 0, \quad \tilde{\Gamma}^i = -\partial_j \tilde{h}^{ij} = 0; \quad (26)$$

negative controls: $h_{x_s y} = G(t, x_s) \Rightarrow \mathcal{M}_y = -\frac{1}{2} \partial_{x_s} \dot{G} \neq 0$; $h_{yy} = h_{zz} = G \Rightarrow \mathcal{H} = -2 \partial_{x_s}^2 G \neq 0$.

Hence the datum (22) sources neither (23) nor (24) at linear order: the three sectors of (22)–(24) commute.

On the boundary wave model of Ref. [?] , $[\partial_t^2 - 2\dot{\beta} \partial_t \partial_x - (1 - \dot{\beta}^2) \partial_x^2] u = 0$, plane-wave reflection [V5]:

$$R_{\text{Sommerfeld}} = -\frac{\dot{\beta}(1 - \dot{\beta})}{(1 + \dot{\beta})(2 - \dot{\beta})} \neq 0, \quad R_{(\partial_t + c_+ \partial_x)} = 0 \quad (\text{identically}). \quad (27)$$

H. Frozen-coefficient boundary stability (node N7)

Laplace–Fourier ansatz $u = \tilde{u} e^{st + \lambda x}$, $\Re s > 0$ [V7]:

$$\lambda_+ = \frac{s}{1 + \dot{\beta}} \text{ (admissible)}, \quad \lambda_- = -\frac{s}{1 - \dot{\beta}}, \quad D_L(s) = \left(s + (1 + \dot{\beta}) \lambda_+ \right)^{L+1} = (2s)^{L+1}, \quad \frac{|D_L(s)|}{|s|^{L+1}} = 2^{L+1}. \quad (28)$$

D_L contains no datum symbol: the injection (22) modifies only the inhomogeneity, so the boundary-stability results of Ref. [?] (constraint sector: all L ; gauge+metric sector: spherical reduction, asymptotically harmonic shift) transfer verbatim with $\|u\| \leq C \|\psi'_0\text{-datum}\|$.

III. THE CHARACTERISTIC SYSTEM

A. Live Cauchy–characteristic matching inside AthenaK (nodes N13–N14)

The ψ_0^{datum} provider of (41) is replaced by a *native, in-process* characteristic solver evolving in lockstep with the Cauchy evolution (one process, no file round-trips). On the compactified radial coordinate $y = 1 - 2r_{\text{wt}}/r$ (Chebyshev–Gauss–Lobatto collocation, LU-factorized global solves, SAT time advance), the solver evolves the *unringed* $\ell=2$ t -scalars (J_t, Q_t, U_t, W_t, H_t) of the linearized Bondi–Sachs hierarchy with the β sources required by live (worldtube-anchored gauge) data — machine-derived as exact-rational fits on the anchored and plain solution families jointly:

$$\begin{aligned} \partial_r(r^2 Q) &= -r^2 \partial_r(\bar{\partial} J) - 4r \bar{\partial} \beta, & \partial_r U &= Q/r^2, \\ \partial_r(r^2 W) &= \frac{1}{2} \bar{\partial} \bar{\partial} J + 2r \bar{\partial} U + \frac{r^2}{2} \partial_r(\bar{\partial} U) - \frac{4}{3} \bar{\partial} \bar{\partial} \beta, & \partial_r(r H) &= \frac{1}{2} r \partial_r^2 J + \partial_r J + \frac{\bar{\partial}^2 \beta}{r} - \frac{\bar{\partial} Q}{2r} - \bar{\partial} U, \end{aligned} \quad (29)$$

with $H = \partial_u J$ and β r -independent at this order. The ringed variables ($rJ, r^2 U, \dots$) diverge at scri on anchored-gauge data ($k = 0, 1$ tails); the unringed system (29) is regular on $y \in [-1, 1]$ in both gauges, and $\psi_0^t = -\partial_r J_t/(2r) - \partial_r^2 J_t/4$ is gauge-invariant (the anchored tails are exactly ψ_0 -silent). Worldtube data are sampled live from the Cauchy ADM state on three geodesic spheres ($\ell=2$ projections; $\partial_t h = -2K$ at linear order) and mapped by a six-row local contract to the anchored boundary scalars $(J, Q, U, W, H, \beta)_{\text{wt}}$. Cone labeling is retarded, $u = t - r$: the boundary datum at Cauchy time t is the ψ_0 probe at r_B on the earlier cone $u = t - r_B$ (causality lag, served from a per-substep history).

Verification. (i) Standalone: anchored end-to-end ψ_0 probe within 8.8×10^{-5} of the exact datum (7.1×10^{-6} at halved step; truncation-dominated). (ii) Live analytic-stub battery: mode-(solver) vs. mode-(self-datum) waveforms agree to 4.1×10^{-9} of peak. (iii) For the purely outgoing test the live datum is conditioning-limited *by physics* ($\psi_0/J \sim 10^{-8}$ peeling: fidelity would need 10^{-11} -relative tube data); the fidelity demonstration instead uses the characteristic-pulse-injection test of Ref. [?] Sec. V.C (ingoing $\ell=2$ pulse on the initial null slice, quiescent Cauchy data, $\psi_0/J = 7 \times 10^{-3}$): the live datum matches the verified solver to 7×10^{-8} of peak in the causally clean window; the transmitted burst exits the Cauchy domain at $t_2 = t_1 + 2R$ within 0.12; a no-datum control shows the entire response is datum-attributable (2.9×10^{-10} of peak) [Fig. 1(a,b)]. (iv) Cross-oracle: SpECTRE CharacteristicExtract (the CCE of Ref. [?]), run on a Bondi–Sachs worldtube file generated from the exact solution, reproduces $\Psi_4 = -\sqrt{6\pi/5} F^{(6)}$ to 6×10^{-6} relative and the datum normalization $\Psi_0 = -\frac{9}{8} F^{(2)}$ (s^2 shape) to 0.1% [Fig. 1(c)]. (v) Constraint rows: a frozen-coefficient LF/GKS analysis of the (Θ, Z_s) boundary realizations shows every matched-falloff advection closure is stable and already absorbs the constraint pair at $1 - O((\sigma/2\omega r)^2)$; falloff-mismatched sink rows self-generate $O(10\%)$ pair reflection — the production default remains the matched (stock) realization of the (23) rows.

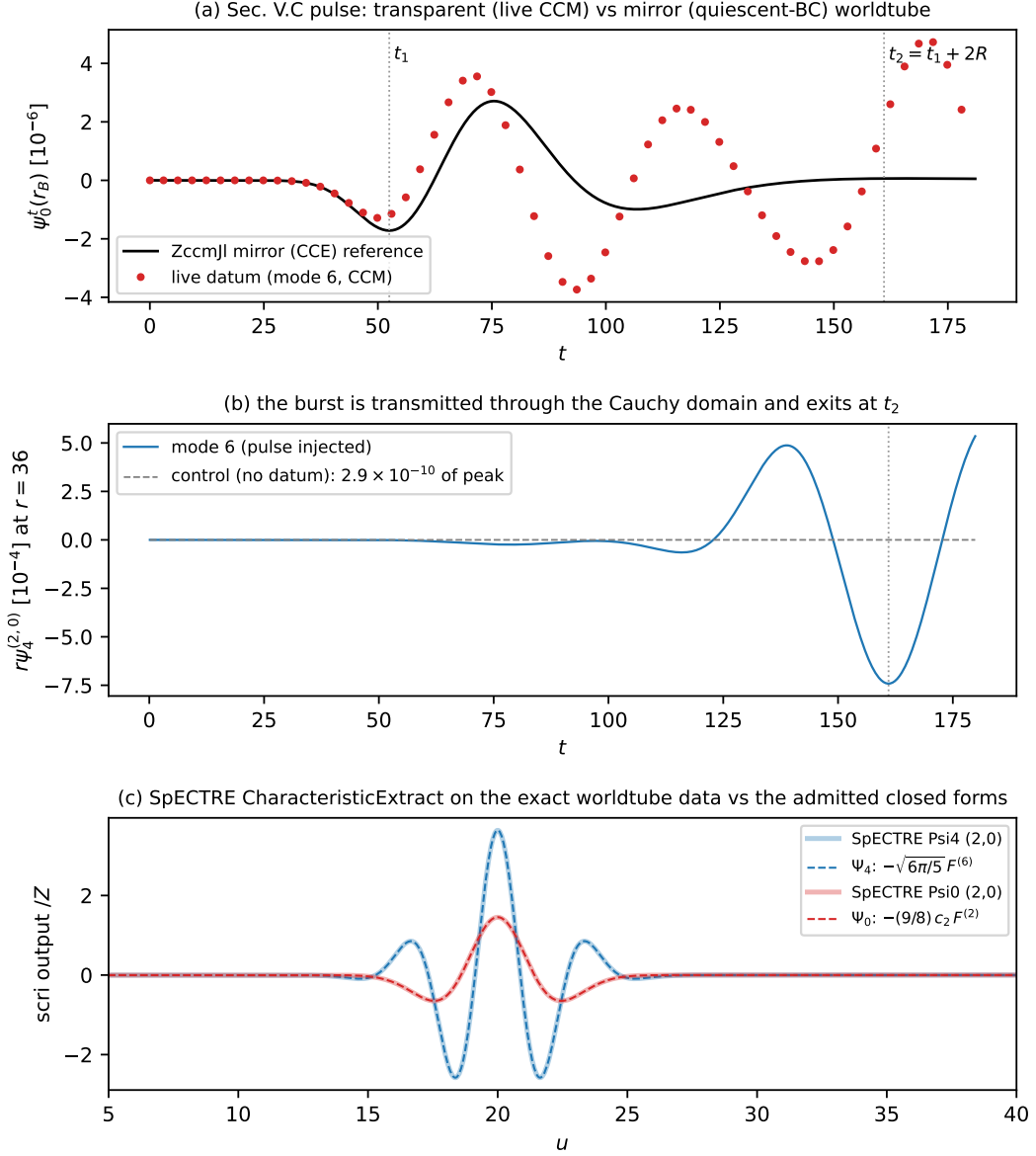


Figure 1. Live-CCM results (Sec. III A). (a) The characteristic-pulse fidelity test: live ψ_0 datum at the Cauchy boundary vs. the quiescent-worldtube (mirror/CCE) reference — the early-window agreement is 7×10^{-8} of peak; the later $O(1)$ separation is the matching physics (the live run transmits and later re-receives the pulse). (b) The transmitted burst at the extraction radius exits at $t_2 = t_1 + 2R$; the no-datum control is consistent with zero. (c) SpECTRE `CharacteristicExtract` on the exact worldtube data vs. the closed forms (shaded wide: oracle output; dashed: $-\sqrt{6\pi/5}F^{(6)}$ and $-\frac{9}{8}c_2F^{(2)}$). Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/n14_native/`, `n14_pulse_ref.csv`, `spectre_oracle/` (ledger nodes N13–N14).

IV. THE MATCHING ALGORITHM

A. Worldtube data map (node N1)

$$g_{00} = -\alpha^2 + \gamma_{ij}\beta^i\beta^j, \quad g_{0i} = \gamma_{ij}\beta^j, \quad g_{ij} = \gamma_{ij}; \quad g^{00} = -\alpha^{-2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2}. \quad (30)$$

ADM kinematics requires $\partial_t \gamma_{ij} = -2\alpha K_{ij} + \beta^k \partial_k \gamma_{ij} + \gamma_{kj} \partial_i \beta^k + \gamma_{ik} \partial_j \beta^k$. From $\gamma_{ij} = \chi^{-1} \tilde{\gamma}_{ij}$,

$$\partial_t \gamma_{ij} = \chi^{-1} \partial_t \tilde{\gamma}_{ij} - \chi^{-2} \tilde{\gamma}_{ij} \partial_t \chi, \quad (31)$$

and inserting (5)–(6) with $K_{ij} = \chi^{-1} (\tilde{A}_{ij} + \frac{1}{3} \tilde{\gamma}_{ij} K)$, $\beta^k \partial_k \gamma_{ij} = \chi^{-1} \beta^k \partial_k \tilde{\gamma}_{ij} - \gamma_{ij} \beta^k \partial_k \ln \chi$,

$$[\partial_t \gamma_{ij}]_{\text{Z4c}} - [\partial_t \gamma_{ij}]_{\text{ADM}} = \gamma_{ij} \left[\frac{2}{3} D_k \beta^k - \frac{2}{3} \partial_k \beta^k + \beta^k \partial_k \ln \chi \right]. \quad (32)$$

$D_i \beta^i = \partial_k \beta^k - \frac{3}{2} \beta^k \partial_k \ln \chi$

[V3: closure exact; naive $D = \partial_k \beta^k$ leaves residual $+1 \cdot \gamma_{ij} \beta^k \partial_k \ln \chi$]. (33)

V. NUMERICAL VERIFICATION

A. Numerical evidence

All runs: JAX float64, $4 \times$ NVIDIA A100-80GB (CUDA_VISIBLE_DEVICES=0,1,2,3), wall clock ≤ 23 s each (budget 600 s); outputs under `results/numerical/`.

Table II. Symbolic and sampling verification of (33)–(28). “Exact” = sympy identity on arbitrary functions or exact rationals (zero float error). Files: `n2_boost_check.txt`, `n3_exact_check.txt`, `n3_dictionary_check.txt`, `n1_varmap_check.txt`, `n4_cpbc_compat_check.txt`, `n5_gauge_check.txt`, `n6_composite_check.txt`, `n7_lf_sketch_check.txt`.

verified statement	exact part	GPU sweep (4 devices)
V1 boost identity (15), factor $\sqrt{2}/2$	12/12 rational points	9.34×10^{-16} over 4,194,304 samples
V2a EXACT identity (19)–(20), $(\sigma, \kappa) = (1, -1)$, $\tilde{\psi}_4$ partner	generic Weyl, all 10 dof	—
V2b linearized on-shell limit (corollary)	arbitrary F_1, F_2	0.0 over 4,194,304 modes (prior form)
V2c Gauss–Codazzi realization (21), Schwarzschild	Codazzi sign +1 derived	E : 7.7×10^{-17} , B : 3.9×10^{-17} , identity 8.5
V3 maps (30), closure (33)	full-symbol identities	round trip 4.10×10^{-16} , gg^{-1} 8.88×10^{-16}
V4 orthogonality (26) + controls	identities and non-vanishing controls	TT residual 0.0 vs control 12.3
V5 reflection (27)	exact R algebra	$ R_{\text{Som}} \geq 0.023$ on $\tilde{\beta} \in [0.05, 0.6]$; $R = 0$ ex
V6 10-mode count, $\psi_0 \rightarrow 0$ reduction, wire-through (exact targets)	structural	reduction 0.0; chain residual 1.47×10^{-16}
V7 Kreiss condition (28), $L = 0..3$	$D_L = (2s)^{L+1}$ exact	ratio deviation 3.2×10^{-14} (4,194,304 sam

Table III. Dynamical model evolution (6th-order interior stencils with a GKS-stable mixed closure — 4th-order Fornberg rows at three edge nodes; fully one-sided 6th-order closures are RK4-unstable (error ledger) — RK4, CFL 0.2, shift $\tilde{\beta} = 0.3$, outgoing Gaussian pulse onto the boundary; one configuration per GPU; `n8_model1d_test.txt`). Reflection is measured in the L_2 (energy) norm — trapezoid sums of Gaussian pulses converge exponentially, so the deviation column is limited by machine arithmetic, not by measurement or truncation. Analytic L_2 relation: $\|q_{\text{in}}\|_{L_2} / \|q_{\text{out}}\|_{L_2} = \frac{\tilde{\beta}}{2+\tilde{\beta}} \sqrt{\frac{1+\tilde{\beta}}{1-\tilde{\beta}}}$. Each row sits at its float64 floor: rows 1–3 at the per-sample roundoff floor (deviation saturates between $N=8193$ and $N=16385$: 1.9 vs 2.1×10^{-14}); the injection row at the linear roundoff-accumulation optimum ($\sim \text{steps} \times \varepsilon$; ladder $N=16385/32769/65537/131073 \Rightarrow 5.4/2.4/5.9/30 \times 10^{-11}$, non-monotone and insensitive to CFL 0.2 vs 0.05).

boundary condition	measured	exact	deviation
Sommerfeld, L_2 ratio ($N=16385$)	0.177752646227	0.177752646227	2.1×10^{-14} (float64 floor)
Sommerfeld, L_2 ratio ($N=8193$)	—	—	1.9×10^{-14} (saturated)
$(\partial_t + c_+ \partial_x)$ [P1/CCM op., $N=16385$]	$R_{L_2} = 1.9 \times 10^{-14}$	0	machine precision
<u>CCM injection (25), rel. L_2 ($N=32769$)</u>	<u>2.4×10^{-11}</u>	<u>0</u>	<u>roundoff-accumulation optimum</u>

B. Dynamical test at the parameters of Ref. [?] (nodes N10–N11)

Full 3+1 evolution: AthenaK Z4c [?] (vendored, pinned; RK4, CFL 0.25, sixth-order stencils at `nghost` = 4, $\kappa_1 = 0.02$), with (22) realized at the Cauchy outer boundary as described below. This subsection is self-contained: every symbol used by the test is defined here or in Secs. II A–II F.

a. Exact solution injected. Even-parity $(\ell, m) = (2, 0)$ Teukolsky wave [?] with the Gaussian profile of Ref. [?], written with physicists' Hermite polynomials H_n and the origin-regular retarded/advanced combinations:

$$F(u) = X e^{-(u-r_c)^2/\tau^2}, \quad F^{(n)}(u) = X \left(-\frac{1}{\tau}\right)^n H_n\left(\frac{u-r_c}{\tau}\right) e^{-(u-r_c)^2/\tau^2}, \quad (34)$$

$$Q^{(n)}(t, r) = F^{(n)}(t-r) - (-1)^n F^{(n)}(t+r), \quad R^{(n)}(t, r) = F^{(n)}(t-r) + (-1)^n F^{(n)}(t+r), \quad \partial_t Q^{(n)} = R^{(n+1)}. \quad (35)$$

Radial functions (regular at $r = 0$ by (35); calligraphic to keep A for the boost (16)):

$$\begin{aligned} \mathcal{A} &= 3 \left[\frac{Q^{(2)}}{r^3} + \frac{3Q^{(1)}}{r^4} + \frac{3Q^{(0)}}{r^5} \right], \quad \mathcal{B} = - \left[\frac{Q^{(3)}}{r^2} + \frac{3Q^{(2)}}{r^3} + \frac{6Q^{(1)}}{r^4} + \frac{6Q^{(0)}}{r^5} \right], \\ \mathcal{C} &= \frac{1}{4} \left[\frac{Q^{(4)}}{r} + \frac{2Q^{(3)}}{r^2} + \frac{9Q^{(2)}}{r^3} + \frac{21Q^{(1)}}{r^4} + \frac{21Q^{(0)}}{r^5} \right], \end{aligned} \quad (36)$$

orthonormal-frame metric perturbation ($\dot{h}_{ij}$ from $Q^{(n)} \rightarrow R^{(n+1)}$ termwise):

$$h_{\hat{r}\hat{r}} = \mathcal{A} \frac{1 + 3 \cos 2\theta}{2}, \quad h_{\hat{r}\hat{\theta}} = -3\mathcal{B} \sin \theta \cos \theta, \quad h_{\hat{\theta}\hat{\theta}} = 3\mathcal{C} \sin^2 \theta - \mathcal{A}, \quad h_{\hat{\phi}\hat{\phi}} = -3\mathcal{C} \sin^2 \theta + \mathcal{A} (3 \sin^2 \theta - 1), \quad (37)$$

$$\gamma_{ij}(0, x) = \delta_{ij} + h_{ij}(0, x), \quad K_{ij}(0, x) = -\frac{1}{2} \dot{h}_{ij}(0, x) \quad (\text{linear order; Ref. [?] solves XCTS}). \quad (38)$$

Weyl scalars of the solution at linear order (the self-datum and the extraction reference):

$$r^5 \psi_0(t, r, \theta) = -\sqrt{\frac{27\pi}{10}} F^{(2)}(t-r) {}_2Y_{20}(\theta), \quad {}_2Y_{20} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \theta, \quad r \psi_4|_{(2,0)} \rightarrow -\sqrt{\frac{6\pi}{5}} F^{(6)}(t-r). \quad (39)$$

b. Discrete realization of (22). At every Cauchy-boundary point, after the CPBC/gauge rows (23)–(24) (Sommerfeld task of Ref. [?]), the two physical rows are driven through $\tilde{A}_{ij}$:

$$s^a = \frac{x^a/r}{\sqrt{\gamma_{bc} x^b x^c}/r}, \quad e_\phi \propto \hat{z} \times s, \quad e_\theta = s \times e_\phi, \quad \bar{m} = \frac{1}{\sqrt{2}} (e_\theta - i e_\phi), \quad (40)$$

$$\partial_t \tilde{A}_{ab}|_{\partial\Omega} = -\frac{2\chi}{\alpha} \left(\psi'_0 \bar{m}_a \bar{m}_b + \bar{\psi}'_0 m_a m_b \right), \quad \psi'_0 = A^2 \psi_0^{\text{datum}}, \quad A = (\alpha - \gamma_{ij} \beta^i s^j) e^{-2\hat{\beta}}$$

(41)

with ψ_0^{datum} the right-hand member of (39) evaluated at the boundary point ($\hat{\beta} = 0$ on the flat background); $\psi_0^{\text{datum}} \rightarrow 0$ reduces (41) to the unmodified scheme exactly (V6; machine-zero reduction in Table III methodology, AthenaK battery: Minkowski preserved to 2.6×10^{-33}).

c. Exact finite-radius reference (node N12). The extraction reference is the *analytic* waveform at the extraction radius itself (no auxiliary reference run): in the AthenaK extraction convention ($= -1 \times$ the scri member of (39) continued to finite r),

$$r \psi_4^{(2,0)}(t, r) = \sqrt{\frac{6\pi}{5}} \left[F^{(6)} + \frac{2F^{(5)}}{r} + \frac{3F^{(4)}}{r^2} + \frac{3F^{(3)}}{r^3} + \frac{3F^{(2)}}{2r^4} \right] (t-r) - \sqrt{\frac{6\pi}{5}} \frac{3F^{(2)}}{2r^4} (t+r), \quad (42)$$

verified by seven symbolic gates (tracelessness, coefficient-exact linearized vacuum, $\sin^2 \theta$ factorization, peeling, scri limit, independent 30-digit finite-difference cross-check at 10^{-40}) and — independently — by the SpECTRE **CharacteristicExtract** cross oracle below.

d. Protocol and result. Parameters of Ref. [?]: $r_c = 20$, $\tau = 2$, $X = 10^{-5}$, Cauchy boundary (worldtube analog) at 41, $t \in [0, 80]$; extraction of $r \psi_4^{(2,0)}$ at $r = 36$; resolution ladder $h = 0.5, 0.331, 0.277$ ($4 \times A100$, every run < 600 s). Error against (42) at each rung's own output times:

$$E(h) = \frac{\|r \psi_4^{(2,0)}[h] - r \psi_4^{(2,0)}[\text{exact}]\|_2}{\|r \psi_4^{(2,0)}[\text{exact}]\|_2} = 4.99 \times 10^{-2}, \quad 5.36 \times 10^{-3}, \quad 2.02 \times 10^{-3}, \quad (43)$$

with measured convergence order 5.39 (first pair) and 5.52 (second pair), finest-rung amplitude $1 \pm 2 \times 10^{-4}$ and time shift 0.000 (Fig. 2). Two production defects were found *only* by this exact-reference program and are corrected in all data shown: a frame-orientation error in the initial data ($\hat{e}_\theta$ flipped; linearized vacuum violated at 5% of $|R_{ij}|$, invisible to constraint monitors and to run-vs-reference protocols since it sat on both sides) and a one-step offset in the waveform time labels. The self-datum CCM–Sommerfeld difference remains peeling-suppressed ($\psi_0 \sim r^{-5}$, relative imprint $\sim 10^{-9}$ of peak at this worldtube), as in Ref. [?] Sec. V.A.

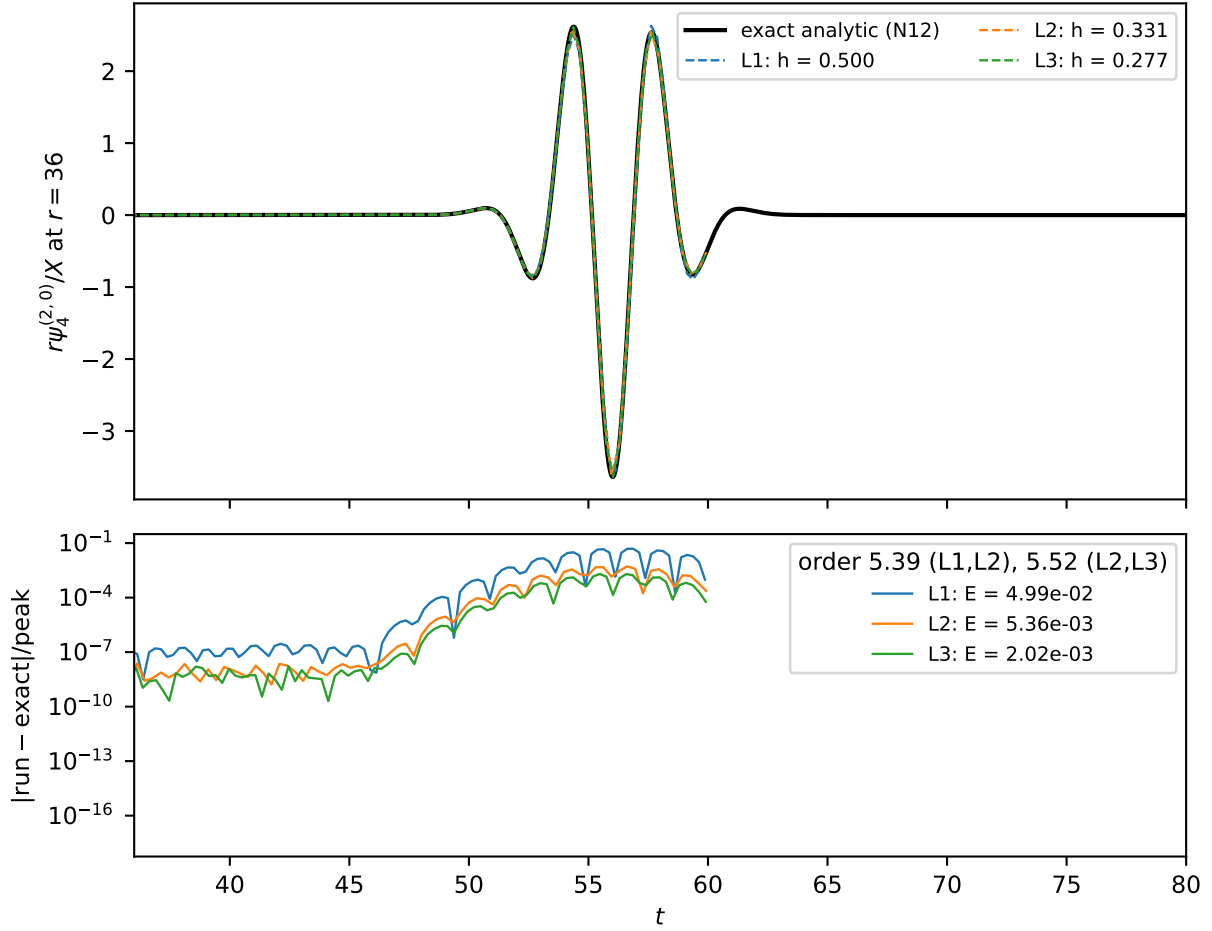


Figure 2. Exact-reference Teukolsky test of Sec. VB ($r_c=20$, $\tau=2$, boundary 41, extraction $r=36$, $X=10^{-5}$). Top: $(\ell, m)=(2, 0)$ mode of $r\psi_4$ — the exact finite-radius waveform (42) (solid) and the three-rung sixth-order ladder. Bottom: pointwise errors against the exact reference; measured order 5.39/5.52, finest-rung relative L_2 error 2.02×10^{-3} (43). Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/athenak_teuk_ana/gpu6/` (ledger node N12).

VI. CONCLUSIONS